

Onyx — The Kits: a developer's guide

Onyx · multi-process OS on Raspberry Pi 4

How an Onyx program is built on the system's shared libraries — the **kits** — with a short example for each. Each kit has its reference, with every operation it exposes (documents 10 to 18, linked in the table below). Companion of the [Developer Guide](#) (the build, the application model, the ports) and of [Kernel Internals](#).

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1. One kit per domain

An Onyx program does not talk to the kernel, and does not carry copies of common code. It is built on **kits**: shared libraries, one per domain, loaded once for the whole system and bound to each program that uses them.

Kit	Domain	On the card	Sources	Reference
AppKit	What makes a program run: the system's calls, strings, the console, <code>.ini</code> files, starting programs	<code>SD:/lib/appkit.so</code>	<code>user/Kits/appkit</code>	10 — AppKit
UIKit	The interface: windows, widgets,	<code>SD:/lib/uikit.so</code>	<code>user/Kits/uikit</code>	11 — UIKit

Kit	Domain	On the card	Sources	Reference
	dialogs, the theme, icons			
SystemKit	Talking to the system and the other programs: notifications, clipboard, trash, volume, wallpaper, file associations, the dock	<code>SD:/lib/systemkit.so</code>	<code>user/Kits/systemkit</code>	12 — SystemKit
NetKit	The network: HTTP, the FTP volumes	<code>SD:/lib/netkit.so</code>	<code>user/Kits/netkit</code>	13 — NetKit
FileKit	Files and folders, paths, compression, archives	<code>SD:/lib/filekit.so</code>	<code>user/Kits/filekit</code>	14 — FileKit
ImageKit	Pictures: reading, writing, resizing, adjusting	<code>SD:/lib/imagekit.so</code>	<code>user/Kits/imagekit</code>	15 — ImageKit
AudioKit	Sound: playing files, notes, synthesis, the sound output	<code>SD:/lib/audiokit.so</code>	<code>user/Kits/audiokit</code>	16 — AudioKit
FontKit	Fonts: FreeType, the font manager, anti-aliased text	<code>SD:/lib/fontkit.so</code>	<code>user/Kits/fontkit</code>	17 — FontKit
PrinterKit	Printing: the Print dialog, a job's pages	<code>SD:/lib/printerkit.so</code>	<code>user/Kits/printerkit</code>	18 — PrinterKit

Three rules follow from this layout, and they hold for every new development:

- **Reusable code goes into the adequate kit** — not into a new shared header, not copied into an application. If no kit fits the domain, a new kit is created (§12).
- **A program draws on the kits as much as it can**, and never reaches the kernel itself. Only AppKit reads the kernel's table; so the kernel can change without any program being rebuilt.
- **What one program alone uses stays beside that program**, in its own folder.

A kit has **one header**, `<kit>/<kit>.h`, which **declares**; the code is in the library. The memory cost of a kit is paid once: its code is shared by every program that uses it.

2. Using a kit in a program

Including

`user/Kits` is on every include path. A source includes a kit's header by the kit's name, whatever its own folder:

```
#include "appkit/appkit.h"      // AppKit
#include "uikit/uikit.h"        // UIKit
#include "systemkit/systemkit.h" // SystemKit
#include "netkit/netkit.h"      // NetKit
#include "filekit/filekit.h"    // FileKit
#include "imagekit/imagekit.h"  // ImageKit
#include "audiokit/audiokit.h"  // AudioKit
#include "fontkit/uikitface.h"  // FontKit
#include "printerkit/printerkit.h" // PrinterKit
```

Linking

AppKit needs nothing: the kernel loads it and binds it to every program. Every other kit is linked through its **import archive**, a few bytes of stubs and a constructor that opens the library before `main`:

The program is	It links
C++ (an application)	<code>lib/<kit>.imp.a</code>
C (a console tool)	<code>lib/<kit>.imp_c.a</code>

An archive only brings what the program calls: listing a kit a program does not use costs nothing, and the library is not opened. The applications' rules in `user/Makefile` already list UIKit, FileKit, SystemKit and NetKit; a console tool names its kits in `user/BinUtils/Makefile`:

```
# a C tool that uses SystemKit
mytool.elf: KITLIBS = ../lib/systemkit.imp_c.a
mytool.elf: ../lib/systemkit.imp_c.a
```

Versions

A kit's table only grows: a function is never removed nor renamed, so a newer library always runs an older program. A program built against a table that grew needs at least that library: its package says so (`needs = systemkit >= 1.56` in `tools/pkg/packages.ini`), and the package manager installs the library first.

From BASIC

A BASIC program uses the kits too: `#import filekit`, then `FileKit.copy (a$, b$, 0, 0)` — every function of a kit that takes and returns numbers, strings and pointers, by its name, and its structures as `TYPEs` (`DIM e AS FileKit.zip_entry`) (docs/04 §13, *The system's kits*). BASIC knows a kit by its **description**, `SD:/lib/<kit>.bi`, made from the kit's `.abi` and its headers by `tools/kitbi/kitbi.py` when the kits are built: nothing in BASIC names a kit, so **a new kit is importable as soon as it is built** (docs/03, *BASIC and the kits*). What BASIC cannot call — C++ classes, structures passed by value, a

structure with a union or a bit field — is left out of the description: a kit meant for BASIC too exposes plain C functions.

On a PC

The same sources build on a PC for the tests and the screenshots. There is no shared library there: the kits' headers bring their code inline, and UIKit's sources are compiled with the program. Nothing changes in the program's code.

3. AppKit — what makes a program run

`#include "appkit/appkit.h"` — nothing to link.

AppKit is the program's link to the system: files, processes, time, windows, sockets... (the `kapi_*` calls, listed in the Developer Guide), plus the small services every program needs: strings without a C library, the console, a reader of `.ini` files, the starting of other programs.

A console tool: read a file, print to the console.

```
#include "appkit/appkit.h"

int main (void)
{
    char buf[256];
    void *f = kapi_open ("SD:/etc/autostart");
    if (f == 0) { ax_putln ("no such file"); return 1; }
    int n = kapi_read (f, buf, sizeof buf - 1);
    kapi_close (f);
    buf[n > 0 ? n : 0] = '\0';

    ax_putln ("SD:/etc/autostart says:");
    ax_putln (buf);
    return 0;
}
```

Settings from an `.ini` file in the program's own folder:

```
if (app_ini_load ("config.ini") >= 0)
{
    const char *name = app_ini_get ("window", "title", "Untitled");
    int width = app_ini_get_int ("window", "width", 640);
    for (int i = 0; i < app_ini_count (); i++) // or every entry, in the file's order
        ax_putln (app_ini_key (i));
}
```

Strings (no C library needed):

```
char path[96]; int n = 0;
ax_strcat (path, sizeof path, &n, "SD:/docs/"); // never overflows, always terminated
ax_strcat (path, sizeof path, &n, "notes.txt");
char num[12]; ax_itoa (42, num);
if (ax_streq (num, "42")) ax_putln (path);
```

Starting a program — an application by its name, or a file by whatever runs it (a `.bas` program by BASIC, a game by its emulator):

```
lx_launch ("tinypad", "SD:/docs/notes.txt");           // an app, with its arguments
lx_open ("SD:/games/tetris.gb", 0);                   // a file: by its runner
```

4. UIKit — the interface

#include "uikit/uikit.h" — link lib/uikit.imp.a (the applications' rules do).

A window is a `Root`; widgets are added to it; `run ()` is the event loop. Everything is in the namespace `uikit`.

A window, a label, a button:

```
#include "appkit/appkit.h"
#include "uikit/uikit.h"
using namespace uikit;

static Label *g_label;
static int g_count;

static void onClick (Widget &)
{
    char text[32]; int n = 0;
    ax_strcat (text, sizeof text, &n, "clicked ");
    char num[12]; ax_itoa (++g_count, num);
    ax_strcat (text, sizeof text, &n, num);
    g_label->setText (text);           // the widget redraws itself
}

int main (void)
{
    Root root (320, 120, "Hello");    // a decorated window
    g_label = new Label (12, 12, 296, 20, "Press the button");
    root.addChild (g_label);
    root.addChild (new Button (12, 44, 100, 28, "Click me", onClick));
    root.run ();                      // until the window is closed
    return 0;
}
```

Dialogs are modal calls:

```
if (uk_messagebox ("Delete", "Delete this file?", MB_YESNO))
{
    char path[256];
    if (uk_file_open (path, sizeof path, "SD:/docs")) // the file chooser
        g_label->setText (path);
}

unsigned colour = 0x2060C0;
if (uk_color_dialog (&colour, "Pick a colour")) root.setBg (colour);
```

The theme: draw with the palette (`C_BG`, `C_TEXT`, `C_ACCENT`, `C_FACE`...), never with fixed colours, so that the program follows the user's theme.

Without C++ (`uikit/flat.h`, in `uikit/uikit.h`): the same window and widgets as handles, behind plain C functions — what a BASIC program calls (`#import UIKit`), what QBStudio's generated code is made of, and what a C program can use:

```
void *win = uk_window ("Hello", 320, 120, 0);
uk_label (win, 12, 12, 200, 20, "Your name:");
void *name = uk_textbox (win, 12, 36, 200, 24, "", 0);
uk_button (win, 220, 36, 80, 24, "OK", on_ok);           /* void on_ok (void *button) */
uk_window_run (win);                                     /* until the window is closed */
```

```
#import UIKit
win = UIKit.window("Hello", 320, 120, 0)
ok = UIKit.button(win, 220, 36, 80, 24, "OK", ADDRESSOF(Clicked))
DO WHILE UIKit.window_wait(win): LOOP
```

A widget added to UIKit that BASIC should reach gets its functions there (a maker, and `uk_set_text` / `uk_get_value` ... taught its kind).

An icon — any picture ImageKit reads (BMP today, PNG tomorrow), given as icons are drawn (magenta = see-through):

```
#include "uikit/bmp.h"

int w, h;
unsigned *px = ui::icon_load ("SD:/apps/tinypad.app/icon.bmp", &w, &h);
if (px) { /* draw it ... */ delete [] px; }
```

A timer or polling: derive from `Root` and override `onTick ()` (called about 60 times a second).

5. SystemKit — talking to the system and the other programs

`#include "systemkit/systemkit.h"` — link `lib/systemkit.imp.a` (C++) or `lib/systemkit.imp_c.a` (C). The one header brings every subject below (a C program gets those written in C: notifications, volume, preload list, the applets' protocol).

A notification (a bubble under the menu bar; `notify_action` adds what a click starts):

```
#include "systemkit/systemkit.h"

notify ("Backup", "3 files copied");
notify_action ("Updates", "2 updates available", "control pkgman");
```

The clipboard, shared by every application:

```
#include "systemkit/systemkit.h"

clip_set_text ("copied text");

char buf[1024];
if (clip_get_text (buf, sizeof buf)) { /* paste buf */ }

int w, h;
unsigned *px = clip_get_image (&w, &h);           // a picture, if one was copied
if (px) { /* ... */ delete [] px; }
```

The trash — a file is moved there instead of being deleted:

```
#include "systemkit/systemkit.h"

if (trash_move ("SD:/docs/old.txt")) notify ("Trash", "old.txt moved to the trash");
int n = trash_count ();                          // what the trash holds
```

Opening a file with the right application (the user's file associations):

```
#include "systemkit/systemkit.h"

char app[32];
if (fa_app_for ("SD:/photos/cat.png", app, sizeof app)) { /* "imageview" */ }
fa_open ("SD:/photos/cat.png");           // a folder, an app, a document: opened
```

The volume, the wallpaper, the dock — a settings program changes the file and tells the system:

```
#include "systemkit/systemkit.h"

volume_save (7, 0);           // 0..10, not muted: kept for the next
start

Wallpaper wp;
wp_load (wp);                 // SD:/etc/wallpaper.ini
/* change wp ... */
wp_save (wp);

dock_reload ();               // the dock reads its settings again
```

Its part (a header of its own, beside systemkit.h)	Subject
notify.h	Notifications
clipboard.h (clippproto.h: its protocol)	The clipboard
trash.h	The trash
fileassoc.h	Which application opens which file
volume.h	The master volume, the output, each program's volume
wallpaper.h	The wallpaper's settings and its painter
dockconf.h	The dock's settings
preloadini.h	The programs loaded ahead at boot
applet_proto.h	A settings applet shown inside the Control Panel

6. NetKit — the network

#include "netkit/netkit.h" — link lib/netkit.imp.a (C++) or lib/netkit.imp_c.a (C). The HttpClient class is apart: #include "netkit/http.hpp".

A page fetched (HTTP/1.0, no allocation: the caller gives the buffer):

```
#include "appkit/appkit.h"
#include "netkit/netkit.h"

static char buf[32 * 1024];

int main (void)
{
    http_response r;
```

```

    if (http_get ("http://example.com/", buf, sizeof buf, &r) < 0 || !r.ok)
    {
        ax_putln ("request failed");
        return 1;
    }
    kapi_stdout_write (r.body, (unsigned) r.body_len);
    return 0;
}

```

A web API (HTTP/1.1, headers, HTTPS with the TLS transport) — the `HttpClient` class:

```

#include "netkit/http.hpp"

static char buf[64 * 1024];
HttpClient http;
http.user_agent ("MyApp/1.0").accept ("application/json");
HttpResponse r = http.get ("http://api.example.com/status", buf, sizeof buf);
if (!r.is_error ()) { /* r.status, the body in buf */ }

```

An FTP server as a volume: once logged in, `FTP:host/dir/file` is a path like any other, for every program.

```

#include "netkit/netkit.h"

if (ftpfs_login ("ftp.example.com", "me", "secret", 1)) // 1: remembered
{
    void *f = kapi_open ("FTP:ftp.example.com/pub/readme.txt");
    /* kapi_read ... kapi_close */
}

```

The raw sockets (`kapi_tcp_connect`, `kapi_tcp_send...`) are AppKit's; NetKit is where protocols built on them belong.

7. FileKit — files, folders, archives

`#include "filekit/filekit.h"` — link `lib/filekit.imp.a`.

A whole file, read and written:

```

#include "filekit/filekit.h"

void *data; unsigned n;
if (fk_load ("SD:/docs/notes.txt", &data, &n) == 0)
{
    /* data: n bytes, a 0 after them */
    fk_save ("SD:/docs/notes.bak", data, n);
    fk_free (data);
}

```

Folders and paths:

```

#include "filekit/filekit.h"

char path[FS_PATHL];
fs_join (path, sizeof path, "SD:/docs", "report.txt");
if (!fs_exists (path)) { /* ... */ }

fk_mkdirs ("SD:/backup/2026/docs"); // every folder of the path

```

```
fk_copy ("SD:/docs", "SD:/backup/2026/docs", 0, 0); // a file, or a folder and all it holds
fk_remove ("SD:/tmp/work"); // the same, removed
```

A ZIP archive, read:

```
char err[80];
fk_zip *z = fk_zip_open ("SD:/downloads/pack.zip", err, sizeof err);
if (z)
{
    struct fk_zip_entry e;
    for (int i = 0; i < fk_zip_count (z); i++)
        if (fk_zip_entry (z, i, &e)) ax_putln (e.name);
    fk_zip_extract_all (z, 0, "SD:/downloads/pack", 0, 0);
    fk_zip_close (z);
}
```

A ZIP archive, written:

```
fk_zipw *w = fk_zipw_create ("SD:/backup/docs.zip");
fk_zipw_add (w, "SD:/docs", 0); // the folder and all it holds
fk_zipw_close (w, 0, 0, err, sizeof err);
```

Other archive formats (tar, tar.gz, gzip) are read through the same calls (`fk_arc_*`); a program asks the library which formats it handles (`fk_arc_formats`) instead of keeping its own list.

8. ImageKit — pictures

`#include "imagekit/imagekit.h"` — link `lib/imagekit.imp.a`.

ImageKit is the system's one picture codec: BMP, GIF, PNG, JPEG, PCX and WebP read; PNG, JPEG, BMP and GIF written.

A picture read, turned, saved in another format:

```
#include "imagekit/imagekit.h"

ik_image *im = ik_load ("SD:/photos/cat.jpg", 0);
if (im)
{
    int w = ik_width (im), h = ik_height (im);
    unsigned *px = ik_pixels (im); // w * h pixels, 0xAARRGGBB
    ik_rotate (im, 1); // a quarter turn clockwise
    ik_save (im, "SD:/photos/cat.png", 90); // the format: the extension's
    ik_image_free (im);
}
```

Pixels of the program's own, saved:

```
ik_image *shot = ik_image_from (pixels, width, height, width);
ik_save (shot, "SD:/screenshots/shot.png", 0);
ik_image_free (shot);
```

The formats, asked — a file chooser's filter follows what the library knows:

```
struct ik_format f[16];
int n = ik_formats (f, 16);
```

Resizing (`ik_scale`), flipping (`ik_flip`), the photo adjustments and the EXIF orientation are in the same header.

9. AudioKit — sound

```
#include "audiokit/audiokit.h" — link lib/audiokit.imp.a.
```

Several programs play at once: the system mixes them, each with its own volume (Control Panel ► Sound).

A sound file (WAV, MP3, OGG, FLAC, MIDI, FM Tracker songs), played in the background:

```
#include "audiokit/audiokit.h"

if (ak_play ("SD:/music/song.mp3", 0) < 0) ax_putln (ak_play_error ());
ak_play_volume (80);                                     // 0..100
/* ... the program goes on ... */
ak_play_stop ();
```

Notes (the General MIDI instruments, a SoundFont):

```
ak_note_on (0, 60, 100);                                // channel, key (60 = middle C),
velocity
kapi_msleep (400);
ak_note_off (0, 60);
```

A sound effect (the FM synthesizer's voices — what the games use):

```
ak_fm_start (0, 440000, 0, 200);                         // voice 0, 440 Hz (in milli-Hz),
volume 0..255
kapi_msleep (120);
ak_fm_stop (0);
```

The program's own samples (an emulator, a synthesizer): stereo, 16 bits, 44.1 kHz (AUDIOKIT_RATE).

```
if (ak_out_open (256, 1024) == 1)
{
    short frames[256 * 2];
    /* fill frames ... */
    ak_out_write (frames, 256);                           // waits for room
    ak_out_close ();
}
```

10. FontKit — fonts and text

```
#include "fontkit/uikitface.h" or "fontkit/fonts.h" — link lib/fontkit.imp.a.
```

FontKit is FreeType for the applications, with the card's fonts (SD:/res/fonts).

Anti-aliased text in a UIKit application — one call before the widgets are built:

```
#include "uikit/uikit.h"
#include "fontkit/uikitface.h"
using namespace uikit;

int main (void)
{
    ft_uikit_install ("DejaVu Sans", 13);                 // every widget's text from now on
    Root root (400, 200, "Smooth text");
    root.addChild (new Label (12, 12, 376, 24, "FreeType draws this"));
    root.run ();
```

```
    return 0;
}
```

The font manager — the families installed, a font at a size, text measured (a word processor, a canvas):

```
#include "fontkit/fonts.h"
```

```
fnt::init ();
for (int i = 0; i < fnt::count (); i++) ax_putln (fnt::name (i));

int fam = fnt::find ("DejaVu Serif");
fnt::Font *f = fnt::get (fam, 0, 14 * 64);           // style 0 (regular), 14 px in 26.6
int width = fnt::advance (f, 'A');                  // in 26.6 units
```

11. PrinterKit — printing

```
#include "printerkit/printerkit.h" — link lib/printerkit.imp.a.
```

A program prints in four steps: the Print dialog, a job, its pages, the end. Lengths are points (1/72 inch) from the page's top-left corner; colours are 0xRRGGBB. The pages are recorded: the print service turns them into what the printer takes — or into a PDF file.

```
#include "printerkit/printerkit.h"
```

```
PrintSetup s;
print_setup_default (&s);
PrintDialogInfo di = { sizeof di, "My document", 1, 0, 0, 0, 0 };
if (print_dialog (&s, &di))           // the printer, the paper, the pages
{
    PrintJob *j = print_begin (&s, "My document");
    int font = print_font (j, "DejaVu Sans", 0);
    if (print_page (j, 0, 0))           // 0, 0: the paper chosen
    {
        print_text (j, font, 12, 72, 84, "Hello, printer", 0x000000);
        print_line (j, 72, 90, s.paper_w - 72, 90, 0.75f, 0x808080);
    }
    print_end (j);                       // to the print queue
}
```

Rectangles, paths (lines and curves, filled or stroked) and pictures are drawn the same way (`print_rect`, `print_path_*`, `print_image`).

12. Adding to a kit, creating a kit

Before writing a helper, look for it in the kits. If two programs could use it, it belongs to a kit.

Adding a function to a kit

1. **Declare** it in the kit's header, with a comment that says what it does.
2. **Write its code** in the kit's source (for SystemKit, NetKit and filekit/fsutil.h: the `.inc` beside the header).
3. **Build:** the kit's table (`<kit>.abi`) gets the new name at its end. Commit the `.abi`. A name is never removed nor renamed: a function that is no longer needed keeps a body.
4. **Raise the package's version** (`tools/pkg/versions.ini: 1.<the table's size>.0`) and what the applications need (`needs = <kit> >= 1.<size>` in `tools/pkg/packages.ini`).

5. **Document** it: its comment in the header is its documentation — run `python tools/docgen/kitdocs.py` (the kit's reference follows), add an example here if it deserves one, then `python docs/build_docs.py`.

Creating a kit

When a domain has no kit (SystemKit and NetKit are the models — small, with no dependency):

- a folder `user/Kits/<kit>/:` **one header a program includes**, `<kit>/<kit>.h` — it may bring a header per subject, each declaring its functions (`XK_API int f (...);`) with a `.inc` beside it for the code — and one source `<kit>.cpp` that compiles the code into the library;
- its rules in `user/Makefile` (copy SystemKit's block): the library `lib/<kit>.so`, its import archives `lib/<kit>.imp.a` (C++) and `lib/<kit>.imp_c.a` (C);
- its package in `tools/pkg/packages.ini` (`required = 1, files = lib/<kit>.so lib/<kit>.bi` — the `.bi` is what BASIC reads of it, made by the build: *From BASIC* above).

The header of a kit a C program may use is written in C.

13. Quick reference

I want to...	Kit	Call
read or write a file	AppKit	<code>kapi_open</code> , <code>kapi_read</code> , <code>kapi_save_file</code>
load a whole file in memory	FileKit	<code>fk_load</code> , <code>fk_save</code>
copy or remove a folder	FileKit	<code>fk_copy</code> , <code>fk_remove</code>
build a path	FileKit	<code>fs_join</code> , <code>fk_path_join</code>
read or write a ZIP	FileKit	<code>fk_zip_open</code> , <code>fk_zipw_create</code>
print to the console	AppKit	<code>ax_puts</code> , <code>ax_putln</code>
read an <code>.ini</code> file	AppKit	<code>app_ini_load</code> , <code>app_ini_get</code>
start another program	AppKit	<code>lx_launch</code> , <code>lx_open</code>
open a window with widgets	UIKit	<code>Root</code> , <code>Label</code> , <code>Button...</code>
ask the user (message, file, colour)	UIKit	<code>uk_messagebox</code> , <code>uk_file_open</code> , <code>uk_color_dialog</code>
load an icon	UIKit	<code>ui::icon_load</code>
show a notification	SystemKit	<code>notify</code> , <code>notify_action</code>
copy / paste	SystemKit	<code>clip_set_text</code> , <code>clip_get_text</code>
move a file to the trash	SystemKit	<code>trash_move</code>
open a file with its application	SystemKit	<code>fa_open</code>
fetch a web page	NetKit	<code>http_get</code> , <code>HttpClient</code>

I want to...	Kit	Call
reach an FTP server	NetKit	<code>ftpfs_login</code> , then <code>FTP:paths</code>
load or save a picture	ImageKit	<code>ik_load</code> , <code>ik_save</code>
resize or turn a picture	ImageKit	<code>ik_scale</code> , <code>ik_rotate</code>
play a sound file	AudioKit	<code>ak_play</code>
play a note or an effect	AudioKit	<code>ak_note_on</code> , <code>ak_fm_start</code>
send samples to the output	AudioKit	<code>ak_out_open</code> , <code>ak_out_write</code>
draw smooth text	FontKit	<code>ft_uikit_install</code> , <code>fnt::get</code>
print	PrinterKit	<code>print_dialog</code> , <code>print_begin</code> , <code>print_page</code> , <code>print_end</code>